

PREDICTION OF CHRONIC KIDNEY DISEASES

*A Mini Project Report submitted in the partial fulfillment of the
Requirements for the award of the degree*

BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING

Submitted by

M. Sai Kumar (18471A0530)
N. Rajesh (18471A0535)
P. John Subhani (18471A0542)
K. S.V. S. Chandra Prasad (18471A0521)

Under the esteemed guidance of

Dr. S. N. Tirumala Rao, M.Tech., Ph.D
Professor & HoD.



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
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NARASARAOPETA ENGINEERING COLLEGE
(AUTONOMOUS)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



CERTIFICATE

This is to certify that the project that is entitled with the name **“PREDICTION OF CHRONIC KIDNEY DISEASES”** is a bonafide work done by the team **M. Sai Kumar (18471A0530), N. Rajesh (18471A0535), P. John Subhani (18471A0542), K. S.V. S. Chandra Prasad (18471A0521)** in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in the Department of **COMPUTER SCIENCE AND ENGINEERING** during 2020-2021.

PROJECT GUIDE

Dr. S. N. Tirumala Rao, M.Tech., Ph.D.,
Professor & HoD

PROJECT CO-ORDINATOR

M. Sireesha, M.Tech.(Ph.D),
Assoc. Professor

HEAD OF THE DEPARTMENT

Dr. S. N. Tirumala Rao, M.Tech., Ph.D.,
Professor & HoD

EXTERNAL EXAMINER

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By

M. Sai Kumar	(18471A0530)
N. Rajesh	(18471A0535)
P. John Subhani	(18471A0542)
K. S.V. S. Chandra Prasad	(18471A0521)

ABSTRACT

Chronic Kidney Disease (CKD) or chronic renal disease has become a major issue with a steady growth rate. A person can only survive without kidneys for an average time of 18 days, which makes a huge demand for a kidney transplant and Dialysis. It is important to have effective methods for early prediction of CKD. Machine learning methods are effective in CKD prediction. This work proposes a workflow to predict CKD status based on clinical data, incorporating data preprocessing, a missing value handling method with collaborative filtering and attributes selection. Out of the 11 machine learning methods considered, the extra tree classifier and random forest classifier are shown to result in the highest accuracy and minimal bias to the attributes. This also considers the practical aspects of data collection and highlights the importance of incorporating domain knowledge when using machine learning for CKD prediction.

INDEX

S. No.	CONTENTS	PAGE NO
1	Introduction	
	1.1 Introduction	1
	1.2 Existing System	1
	1.3 Proposed System	1
	1.4 System Requirements	2
2	Literature Survey	
	2.1 Machine Learning	3
	2.2 Some machine learning methods	3
	2.3 Applications of machine learning	4
	2.4 Prevalence of chronic kidney diseases	4
	2.5 Importance of machine learning in healthcare	5
	2.6 Classification	6
	2.7 Implementation of machine learning using python	7
	2.8 Need of Data Preprocessing	10
	2.9 Machine Learning Products	11
3	System Analysis	
	3.1 Scope of the project	13
	3.2 Analysis	13
	3.3 Data Preprocessing	14
	3.4 Confusion Matrix	18
4	Design	21
5	Implementation	22
6	Result Analysis	30
7	Screenshots	31
8	Conclusion	34
9	Future Scope	35
10	References	36

LIST OF FIGURES

S.NO.	LIST OF FIGURES	PAGE NO
1	Fig 2.4.1: graph showing various stages of CKD and its growth	5
2	Fig 2.8.1: Need of Data Preprocessing	10
3	Fig 3.2.1: Dataset of CKD	14
4	Fig 3.3.1: Co-relation matrix for CKD	16
5	Fig 3.3.2: class labels and attributes	17
6	Fig 3.4.1: format of confusion matrix	19
7	Fig 4.1: modeling the system	21
8	Fig 6.1: Accuracy after applying PCA and Correlation	30
9	Fig.7.1: Output screen for Home page	31
10	Fig 7.2: Report based on risk result of basic checkup-1	31
11	Fig 7.3: Prediction form	32
12	Fig 7.4: Predicted form of No CKD	32
13	Fig 7.5: Predicted form of CKD	33
14	Fig 7.6: Report based on no risk result of basic checkup-2	33

1. INTRODUCTION

1.1 Introduction

Machine learning is one of the applications of artificial intelligence (AI) that provides computers, the ability to learn automatically and improve from experience instead of explicitly programmed. It focuses on developing computer programs that can access data and use it to learn from themselves. The main aim is to allow computers to learn automatically without human intervention and also adjust actions accordingly. Chronic Kidney diseases had a great deal of attention in medical research. The diagnosis of kidney disease is a challenging task, which can offer automated prediction about the heart condition of patient so that further treatment can be made effective.

1.2 Existing System

Hospitals maintain all the patient records. Even though, those records are not used in an efficient manner for diagnosis. To maintain the records in an efficient error free manner, the new proposed system is introduced.

Disadvantages:

1. Doesn't generate accurate and efficient results
2. Computation time is very high
3. Difficulty in maintenance of patient records
4. Lacking of accuracy may result in lack of efficient further treatment

1.3 Proposed System

We proposed to develop a system which will help practitioners to predict kidney related disease based on some attributes like age, gender, blood pressure and so on. So, there is a need for developing a decision system which will help practitioners to predict the kidney disease in an easier way, which can offer prediction about the kidney condition of patient so that further treatment can be made effectively. This proposed system not only accurately predicts kidney disease but also reduces time for prediction. The machine learning algorithms like decision tree, random forest, Naïve Bayes, KNearestNeighbours have proven to be most accurate & reliable and hence used in this project.

Advantages:

1. Generates accurate and efficient results
2. Computation time is greatly reduced
3. Easy maintenance of patient records
4. Reduces manual work
5. Efficient further treatment
6. Automated prediction

1.4 System Requirements**Hardware Requirements:**

- SystemType : Intel Core i3 or above
- Cachememory : 4MB(Megabyte)
- RAM : 8 gigabyte (GB)
- BusSpeed : 5 GT/s DBI2
- Number of cores : 2
- Number of threads : 4

Software Requirements:

- Operating System : Windows 10 Home, 64 bit Operating System
- Coding Language : Python
- Python distribution : Anaconda, Spyder

2. LITERATURE SURVEY

2.1 Machine Learning

Machine learning is one of the applications of artificial intelligence (AI) that provides computers, the ability to learn automatically and improve from experience instead of explicitly programmed. It focuses on developing computer programs that can access data and use it to learn from themselves. The main aim is to allow computers to learn automatically without human intervention and also adjust actions accordingly.

2.2 Some machine learning methods

Machine learning algorithms are often categorized as supervised and unsupervised.

- **Supervised machine learning algorithms** can apply what has been learned in the past to new data using labeled examples to predict future events. Starting from the analysis of a known training dataset, the learning algorithm produces an inferred function to make predictions about the output values. The system is able to provide targets for any new input after sufficient training. The learning algorithm can also compare its output with the correct, intended output and find errors in order to modify the model accordingly.
- In contrast, **unsupervised machine learning algorithms** are used when the information used to train is neither classified nor labeled. Unsupervised learning studies how systems can infer a function to describe a hidden structure from unlabeled data. The system doesn't figure out the right output, but it explores the data and can draw inferences from datasets to describe hidden structures from unlabeled data.
- **Semi-supervised machine learning algorithms** fall somewhere in between supervised and unsupervised learning, since they use both labeled and unlabeled data for training – typically a small amount of labeled data and a large amount of unlabeled data. The systems that use this method are able to considerably improve learning accuracy. Usually, semi-supervised learning is

chosen when the acquired labeled data requires skilled and relevant resources in order to train it / learn from it. Otherwise, acquiring unlabeled data generally doesn't require additional resources.

- **Reinforcement machine learning algorithms** is a learning method that interacts with its environment by producing actions and discovers errors or rewards. Trial and error search and delayed reward are the most relevant characteristics of reinforcement learning. This method allows machines and software agents to automatically determine the ideal behaviour within a specific context in order to maximize its performance. Simple reward feedback is required for the agent to learn which action is best. This is known as there inforcement signal.

2.3 Applications of machine learning

1. Virtual Personal Assistants
2. Predictions while Commuting
3. Videos Surveillance
4. Social Media Services
5. Email Spam and Malware Filtering
6. Online Customer Support
7. Search Engine Result Refining
8. Product Recommendations
9. Online Fraud Detection

2.4 Prevalence of chronic kidney diseases

Chronic kidney disease is a major public health problem related to serious sequelae, including end-stage kidney failure, cardiovascular disease, premature death, and increased health care expenditures. The risk for these serious sequelae increases as the glomerular filtration rate (GFR) decreases below 60 ml/minute/1.73 m corresponding to the National Kidney Foundation's Kidney Disease Outcomes Quality Initiative (K/DOQI) stage 3 and stage 4 chronic kidney disease defined as GFR 15 to 59ml/minute/1.73 m² Fewer than 25% of the estimated seven million adults with chronic kidney disease reported ever being told that they had weak or failing kidneys, suggesting that more than 75% of adults with chronic kidney disease are undiagnosed in the United States (U.S.).

The identification of high-risk individuals provides an opportunity for early detection and intervention to prevent or delay the onset of end-stage renal disease, cardiovascular events, premature death, and increased health care expenditures. Traditional risk factors

for chronic kidney disease include age >60 years, hypertension, diabetes, poor glycemic control, obesity, macro albuminuria, smoking, C-reactive protein (CRP), elevated total cholesterol, low levels of high -density lipoprotein (HDL) cholesterol, elevated levels of low- density lipoprotein cholesterol, race/ethnicity, gender, and income/poverty. Non- traditional risk factors that may contribute to chronic kidney disease include periodontal disease, education, and access to medical care.

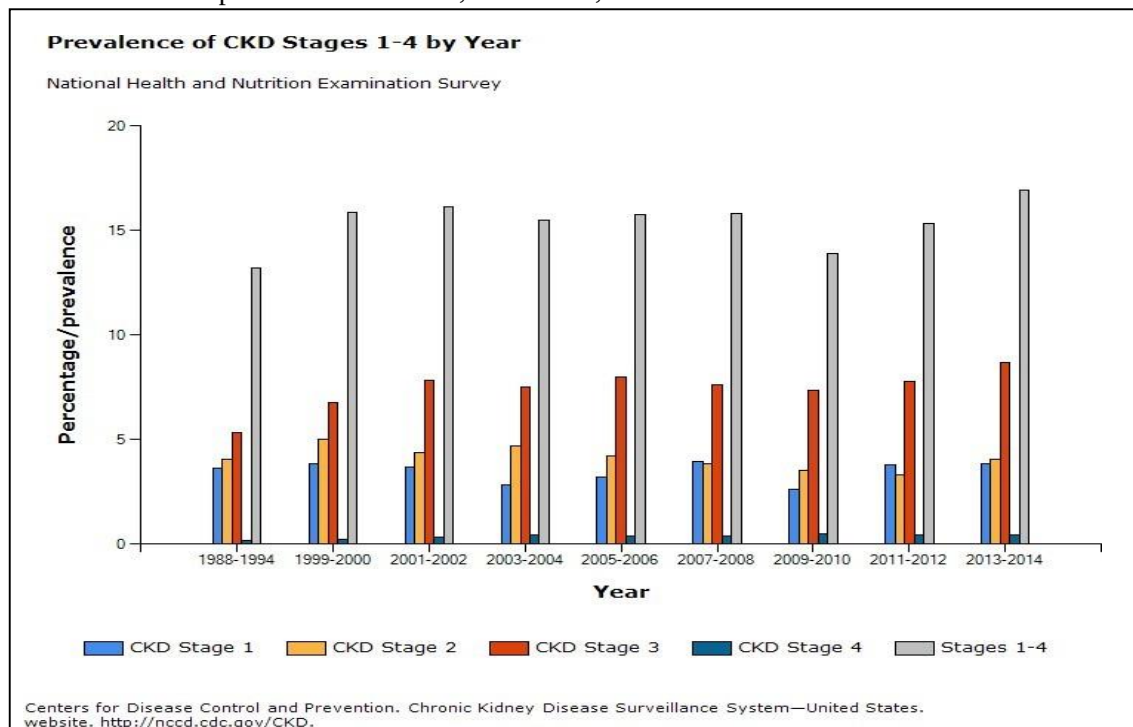


Fig 2.4.1: graph showing various stages of CKD and its growth

From the above graph we can say that year by year the death rate due to chronic kidney disease is increasing compare to other diseases.

2.5 Importance of machine learning in healthcare

The importance of machine learning in healthcare is increasing because of its ability to process huge datasets efficiently beyond the range of human capability, and then dependably convert analysis of that data into clinical insights that assist physicians in planning and providing care, which ultimately leads to better outcomes, reduces the costs of care, and increases patients satisfaction. Using these types of advanced analytics, we can provide better information to doctors at the point of patient care.

2.6 Classification

- It is a process of categorizing data into given classes. Its primary goal is to identify the class of our new data.

Machine learning algorithms for classification

Research on data mining has led to the formulation of several data mining algorithms. These algorithms can be directly used on a dataset for creating some models or to draw vital conclusions and inferences from that dataset. Some popular data mining algorithms are Decision tree, Naïve Bayes, k-means, artificial neural network etc.

Decision Tree:

Decision Tree Analysis is a general, predictive modelling tool that has applications spanning a number of different areas. In general, decision trees are constructed via an algorithmic approach that identifies ways to split a data set based on different conditions. It is one of the most widely used and practical methods for supervised learning. Decision Trees are a non-parametric supervised learning method used for both classification and regression tasks. The goal is to create a model that predicts the value of a target variable by learning simple decision rules inferred from the data features.

The decision rules are generally in form of if-then-else statements. The deeper the tree, the more complex the rules and fitter the model.

1. Naive Bayes (NB): It is a simple technique for constructing classifiers. It is a probabilistic classifier based on Bayes' theorem. All Naive Bayes classifiers assume that the value of any particular feature is independent of the value of any other feature, given the class variable. Bayes theorem is given as follows: $P(C|X) = P(X|C) * P(C)/P(X)$, where X is the data tuple and C is the class such that $P(X)$ is constant for all classes. Though it assumes an unrealistic condition that attribute values are conditionally independent, it performs surprisingly well on large datasets where this condition is assumed and holds.

2. Random Forest: Random Forests are an ensemble learning method (also thought of as a form of nearest neighbour predictor) for classification and regression techniques. It builds multiple decision trees and then merges them together in-order to get more accurate and stable predictions. It constructs a number of Decision trees at training time and outputs the class that is the mode of the classes output by individual trees. It also tries to minimize the problems of high variance and high bias by averaging to find a natural balance between the two extremes. Both R and Python have robust packages to implement this algorithm.

3. KNN: KNN algorithm is one of the simplest classification algorithms and it is one of the most used learning algorithms. KNN is a non-parametric, lazy learning algorithm. Its purpose is to use a dataset in which the data points are separated into several classes to predict the classification of a new sample point. A KNN algorithm uses a data and classifies new data points based on a similarity measures (e.g. distance function, error rate). Classification is done by a majority vote to its neighbours. The data is assigned to the class which has the most nearest neighbours. As we increase the number of nearest neighbours, the value of k , accuracy may increase.

When we say a technique is non-parametric, it means that it does not make any assumptions on the underlying data distribution. In other words, the model structure is determined from the data. If you think about it, it's pretty useful, because in the "real world", most of the data does not obey the typical theoretical assumptions made (as in linear regression models, for example). Therefore, KNN could and probably should be one of their choices for a classification study when there is little or no prior knowledge about the distribution data.

4. Logistic Regression: Logistic regression is one of the most popular Machine Learning algorithms, which comes under the Supervised Learning technique. It is used for predicting the categorical dependent variable using a given set of independent variables. Logistic regression predicts the output of a categorical dependent variable. Therefore the outcome must be a categorical or discrete value. It can be either Yes or No, 0 or 1, true or False, etc. but instead of giving the exact value as 0 and 1, it gives the probabilistic values which lie between 0 and 1. Logistic Regression is much similar to the Linear Regression except that how they are used. Linear Regression is used for solving Regression problems, whereas Logistic regression is used for solving the classification problems.

2.7 Implementation of machine learning using Python

Python is a popular programming language. It was created in 1991 by Guido van Rossum. It is used for:

- ☐ Web Development (server-side),
- ☐ Software Development,
- ☐ Mathematics,
- ☐ system scripting

The most recent major version of Python is Python 3. However, Python 2, although not being updated with anything other than security updates, is still quite popular.

It is possible to write Python in an Integrated Development Environment, such as Thonny, Pycharm, Netbeans or Eclipse, Anaconda which are particularly useful when managing larger collections of Python files. Python was designed for its readability. Python

uses new lines to complete a command, as opposed to other programming languages which often use semicolons or parentheses. Python relies on indentation, using whitespace, to define scope; such as the scope of loops, functions and classes. Other programming languages often use curly-brackets for this purpose. In the older days, people used to perform Machine Learning tasks manually by coding all the algorithms and mathematical and statistical formula. This made the process time consuming, tedious and inefficient. But in the modern days, it is become very much easy and efficient compared to the olden days by various python libraries, frameworks, and modules. Today, Python is one of the most popular programming

languages for this task and it has replaced many languages in the industry, one of the reason is its vast collection of libraries. Python libraries that used in Machine Learning are:

- Numpy
- Scipy
- Scikit-learn
- Theano
- TensorFlow
- Keras
- PyTorch
- Pandas
- Matplotlib

NumPy: is a very popular python library for large multi-dimensional array and matrix processing, with the help of a large collection of high-level mathematical functions. It is very useful for fundamental scientific computations in Machine Learning. It is particularly useful for linear algebra, Fourier transform, and random number capabilities. High-end libraries like TensorFlow uses NumPy internally for manipulation of Tensors.

SciPy: is a very popular library among Machine Learning enthusiasts as it contains different modules for optimization, linear algebra, integration and statistics. There is a difference between the SciPy library and the SciPy stack. The SciPy is one of the core packages that make up the SciPy stack. SciPy is also very useful for image manipulation.

Scikit-learn: is one of the most popular Machine Learning libraries for classical Machine Learning algorithms. It is built on top of two basic Python libraries, NumPy and SciPy. Scikit-learn supports most of the supervised and unsupervised learning algorithms. Scikit-learn can also be used for data-mining and data-analysis, which makes it a great tool who is starting out with Machine Learning.

Theano: is a popular python library that is used to define, evaluate and optimize mathematical expressions involving multi-dimensional arrays in an efficient manner. It is achieved by optimizing the utilization of CPU and GPU. It is extensively used for unit-testing and self-verification to detect and diagnose different types of errors. Theano is a very powerful library that has been used in large-scale computationally intensive scientific projects for a long time but is simple and approachable enough to be used by individuals for their own projects.

TensorFlow: is a very popular open-source library for high performance numerical computation developed by the Google Brain team in Google. As the name suggests, Tensorflow is a framework that involves defining and running computations involving tensors. It can train and run deep neural networks that can be used to develop several AI applications. TensorFlow is widely used in the field of deep learning research and application.

Keras: is a very popular Machine Learning library for Python. It is a high -level neural networks API capable of running on top of TensorFlow, CNTK, or Theano. It can run seamlessly on both CPU and GPU. Keras makes it really for ML beginners to build and design a Neural Network. One of the best thing about Keras is that it allows for easy and fast prototyping.

PyTorch: is a popular open-source Machine Learning library for Python based on Torch, which is an open-source Machine Learning library which is implemented in C with a wrapper in Lua. It has an extensive choice of tools and libraries that supports on Computer Vision, Natural Language Processing(NLP) and many more ML programs. It allows developers to perform computations on Tensors with GPU acceleration and also helps in creating computational graphs.

Pandas: is a popular Python library for data analysis. It is not directly related to Machine Learning. As we know that the dataset must be prepared before training. In this case, Pandas comes handy as it was developed specifically for data extraction and preparation. It provides high-level data structures and wide variety tools for data analysis. It provides many inbuilt methods for groping, combining and filtering data.

Matplotlib: is a very popular Python library for data visualization. Like Pandas, it is not directly related to Machine Learning. It particularly comes in handy when a programmer wants to visualize the patterns in the data. It is a 2D plotting library used for creating 2D graphs and plots. A module named pyplot makes it easy for programmers for plotting as it provides features to control line styles, font properties, formatting axes, etc. It provides various kinds of graphs and plots for data visualization, histogram, errorcharts, barcharts, etc.

2.8 Need of Data Pre-processing

Pre-processing refers to the transformations applied to our data before feeding it to the algorithm. Data Pre-processing is a technique that is used to convert the raw data into a clean dataset. In other words, whenever the data is gathered from different sources it is collected in raw format which is not feasible for the analysis.

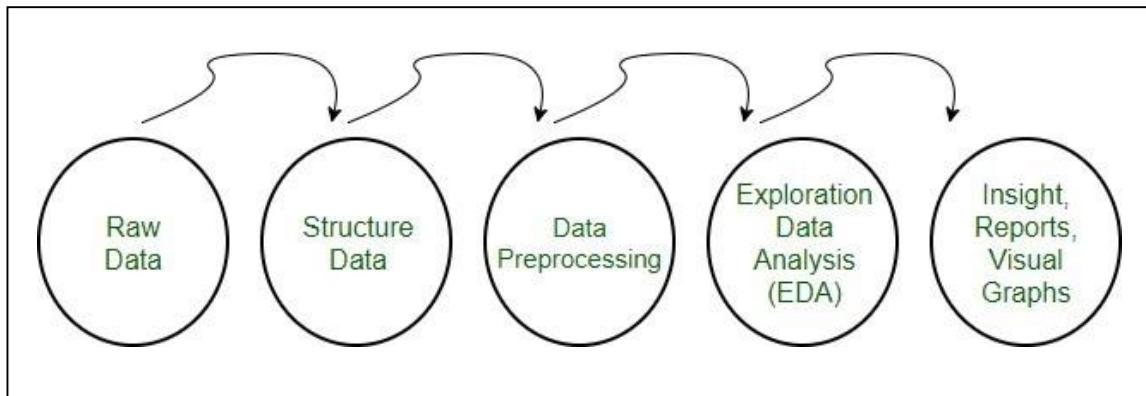


Fig 2.8.1: Need of Data Preprocessing

For achieving better results from the applied model in Machine Learning projects the format of the data has to be in a proper manner. Some specified Machine Learning model needs information in a specified format. For example, Random Forest algorithm does not support null values, therefore to execute random forest algorithm null values have to be managed from the original raw data set.

Another aspect is that dataset should be formatted in such a way that more than one Machine Learning and Deep Learning algorithms are executed in one dataset, and best out of them is chosen.

2.9 Machine learning products

Data mining is widely used in the medical field such as prediction of chronic kidney disease since it is a multidisciplinary field. Using data mining researchers are developing various techniques in order to predict the chronic kidney diseases with high accuracy. Large no. of research work is carried out for medical diagnosis for various diseases

Prediction of chronic kidney disease using k-nearest neighbor and particle swarm optimization was introduced Jabbar MA[3]. Feature subset selection is used to solve this problem. Feature selection improved accuracy and reduced the running time. Before feature subset selection accuracy obtained is 72.4%. PSO search filters the number of features and selects the features which contribute more to the classification. By applying KNN with PSO accuracy improved to 92.7%.

S.Dilli Arasu and Dr. R. Thirumalaiselvi [3] has worked on missing values in a dataset of chronic Kidney Disease. Missing values in dataset will reduce the accuracy of our model as well as prediction results. They find solution over this problem that they performed a recalculation process on CKD stages and by doing so they got up with unknown values. They replaced missing values with recalculated values.

Asif salekin and john stankovic [7] they use novel approach to detect CKD using machine learning algorithm. They get result on dataset which having 400 records and 25 attributes which gives result of patient having CKD or not CKD. They use k-nearest neighbours, random forest and neural network to get results. For feature reduction they use wrapper method which detect CKD with high accuracy.

Pinar Yildirim [8] searches the effect of class imbalance when we train the data by using development of neural network algorithm for making medical decision on chronic kidney disease. In this proposed work, a comparative study was performed using sampling algorithm. This study reveals that the performance of classification algorithms can be improved by using the sampling algorithms. It also reveals that the learning rate is a crucial parameter which significantly effect on multilayer perceptron.

Sahil Sharma, Vinod Sharma, and Atul Sharma [9], has assessed 12 different classification algorithm on dataset which having 400 records and 24 attributes. They had compared their calculated results with actual results for calculating the accuracy of prediction results. They used assessment metrics like accuracy, sensitivity, precision and specificity. They find that the decision tree technique gives accuracy up to 98.6%, sensitivity of 0.9720, and precision of 1 and

specificity of 1.

Gopika and Vanitha [2] employ a clustering technique for accurate CKD detection and reduced diagnosis time. They use fuzzy c-means, k-means, and k-medoids techniques and show 87% accuracy using the fuzzy c-means clustering technique for a dataset from the UCI machine repository.

Charleonnann et al. [5] employ decision tree, logistic regression, support vector machine (SVM), and k-nearest neighbor (KNN) as classifiers for CKD detection using a dataset with 2 classes, 400 instances, and 24 attributes. They use a CKD dataset from the UCI machine learning repository. The results show the SVM technique as the better detection technique for detection accuracy and sensitivity.

Padmanaban and Parthiban [8] explore early kidney disease detection using machine learning techniques and a dataset of 600 instances to validate decision tree and naive Bayes, achieving 91% detection accuracy through the decision tree and reporting 95% sensitivity and 94% specificity with the decision tree for CKD detection.

Wibawa et al. [11] use feature selection and boosted classifier to diagnose CKD and employ AdaBoost for ensemble learning and correlation-based feature selection (CFS).

They use naive Bayes, k-nearest neighbor (KNN), support vector

machine for CKD detection and conclude AdaBoost and CFS as the most promising classifiers in addition KNN and naive Bayes classifiers in CKD detection. They achieve 0.98 f-measure rate, 0.98 recall rate, and 0.981 accuracy rate.

Wickramasinghe et al. [12] control CKD using an appropriate diet plan and recommend diet plans to different patients through their classification method. They recommend a diet plan based on the patient's blood potassium level. They use multicast neural network, multiclass decision jungle, multiclass logistic regression, multiclass decision forest and achieve 99.17% accuracy through the multiclass decision forest algorithm

3. SYSTEM ANALYSIS

3.1 Scope of the project

The scope of this system is to maintain patient details in datasets, train the model using the large quantity of data present in datasets and predict whether presence or absence of disease on new data during testing.

3.2 Analysis

The dataset that we use for the prediction of chronic kidney disease was taken from the UCI repository which contain the details of following attributes to train and test the system for the prediction

1. age –age
2. bp - bloodpressure
3. sg - specificgravity
4. al –albumin
5. su–sugar
6. rbc - red bloodcells
7. pc - pus cell
8. pcc - pus cellclumps
9. ba –bacteria
10. bgr- blood glucoserandom
11. bu - bloodurea
12. sc - serumcreatinine
13. sod –sodium
14. pot –potassium
15. hemo – hemoglobin
16. pcv - packed cellvolume

17. wc - white blood cellcount
18. rc - red blood cellcount
19. htn –hypertension
20. dm - diabetesmellitus
21. cad - coronary artery disease
22. appet –appetite
23. pe - pedaledema
24. ane –anemia
25. class -class

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y
1	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr	bu	sc	sod	pot	hemo	pcv	wbcc	rbcc	htn	dm	cad	appet	pe	ane	classification
2	48	80	1.02	1	0		normal	notpreser	notpreser	121	36	1.2			15.4	44	7800	5.2	yes	yes	no	good	no	no	ckd
3	7	50	1.02	4	0		normal	notpreser	notpreser		18	0.8			11.3	38	6000		no	no	no	good	no	no	ckd
4	62	80	1.01	2	3	normal	normal	notpreser	notpreser	423	53	1.8			9.6	31	7500		no	yes	no	poor	no	yes	ckd
5	48	70	1.005	4	0	normal	abnormal	present	notpreser	117	56	3.8	111	2.5	11.2	32	6700	3.9	yes	no	no	poor	yes	yes	ckd
6	51	80	1.01	2	0	normal	normal	notpreser	notpreser	106	26	1.4			11.6	35	7300	4.6	no	no	no	good	no	no	notckd
7	60	90	1.015	3	0			notpreser	notpreser	74	25	1.1	142	3.2	12.2	39	7800	4.4	yes	yes	no	good	yes	no	ckd
8	68	70	1.01	0	0		normal	notpreser	notpreser	100	54	24	104	4	12.4	36			no	no	no	good	no	no	notckd
9	24		1.015	2	4	normal	abnormal	notpreser	notpreser	410	31	1.1			12.4	44	6900	5	no	yes	no	good	yes	no	notckd
10	52	100	1.015	3	0	normal	abnormal	present	notpreser	138	60	1.9			10.8	33	9600	4	yes	yes	no	good	no	yes	ckd
11	53	90	1.02	2	0	abnormal	abnormal	present	notpreser	70	107	7.2	114	3.7	9.5	29	12100	3.7	yes	yes	no	poor	no	yes	notckd
12	50	60	1.01	2	4		abnormal	present	notpreser	490	55	4			9.4	28			yes	yes	no	good	no	yes	ckd
13	63	70	1.01	3	0	abnormal	abnormal	present	notpreser	380	60	2.7	131	4.2	10.8	32	4500	3.8	yes	yes	no	poor	yes	no	notckd
14	68	70	1.015	3	1		normal	present	notpreser	208	72	2.1	138	5.8	9.7	28	12200	3.4	yes	yes	yes	poor	yes	no	ckd
15	68	70						notpreser	notpreser	98	86	4.6	135	3.4	9.8				yes	yes	yes	poor	yes	no	notckd
16	68	80	1.01	3	2	normal	abnormal	present	present	157	90	4.1	130	6.4	5.6	16	11000	2.6	yes	yes	yes	poor	yes	no	ckd
17	40	80	1.015	3	0		normal	notpreser	notpreser	76	162	9.6	141	4.9	7.6	24	3800	2.8	yes	no	no	good	no	yes	ckd
18	47	70	1.015	2	0		normal	notpreser	notpreser	99	46	2.2	138	4.1	12.6				no	no	no	good	no	no	ckd
19	47	80						notpreser	notpreser	114	87	5.2	139	3.7	12.1				yes	no	no	poor	no	no	notckd
20	60	100	1.025	0	3		normal	notpreser	notpreser	263	27	1.3	135	4.3	12.7	37	11400	4.3	yes	yes	yes	good	no	no	ckd
21	62	60	1.015	1	0		abnormal	present	notpreser	100	31	1.6			10.3	30	5300	3.7	yes	no	yes	good	no	no	ckd

Fig 3.2.1: Dataset of CKD

3.3 Data Pre-processing

Before feeding data to an algorithm we have to apply transformations to our data which is referred as pre-processing. By performing pre-processing the raw data which is not feasible for analysis is converted into clean data. In-order to achieve better results using a model in Machine Learning, data format has to be in a proper manner. The data should be in a particular format for different algorithms. For example, if we consider Random

Forest algorithm it does not support null values. So that those null values have to be managed using raw data.

Missing values

Filling missing values is one of the pre-processing techniques. The missing values in the dataset is represented as '?' but it a non-standard missing value and it has to be converted into a standard missing value NaN. So that pandas can detect the missing values. The fig1 below is a heatmap representing the missing values. In these graph missing values are present in ca, thal features. We have filled that missing values using the median of the features. After filling the missing values our heat map looks like

Column name	Missing values counts
id	0
age	9
bp	12
sg	4746
al	46
su	49
rbc	152
pc	65
pcc	4
Ba	4
bgr	44
Bu	19
sc	17
sod	87
pot	88
hemo	52
pvc	72
wc	105
rc	130
htn	5
dm	2
cad	2
appet	1

```

pe          1
ane        1
classification 0

```

```
df = pd.read_csv('mycsvdata.csv')
```

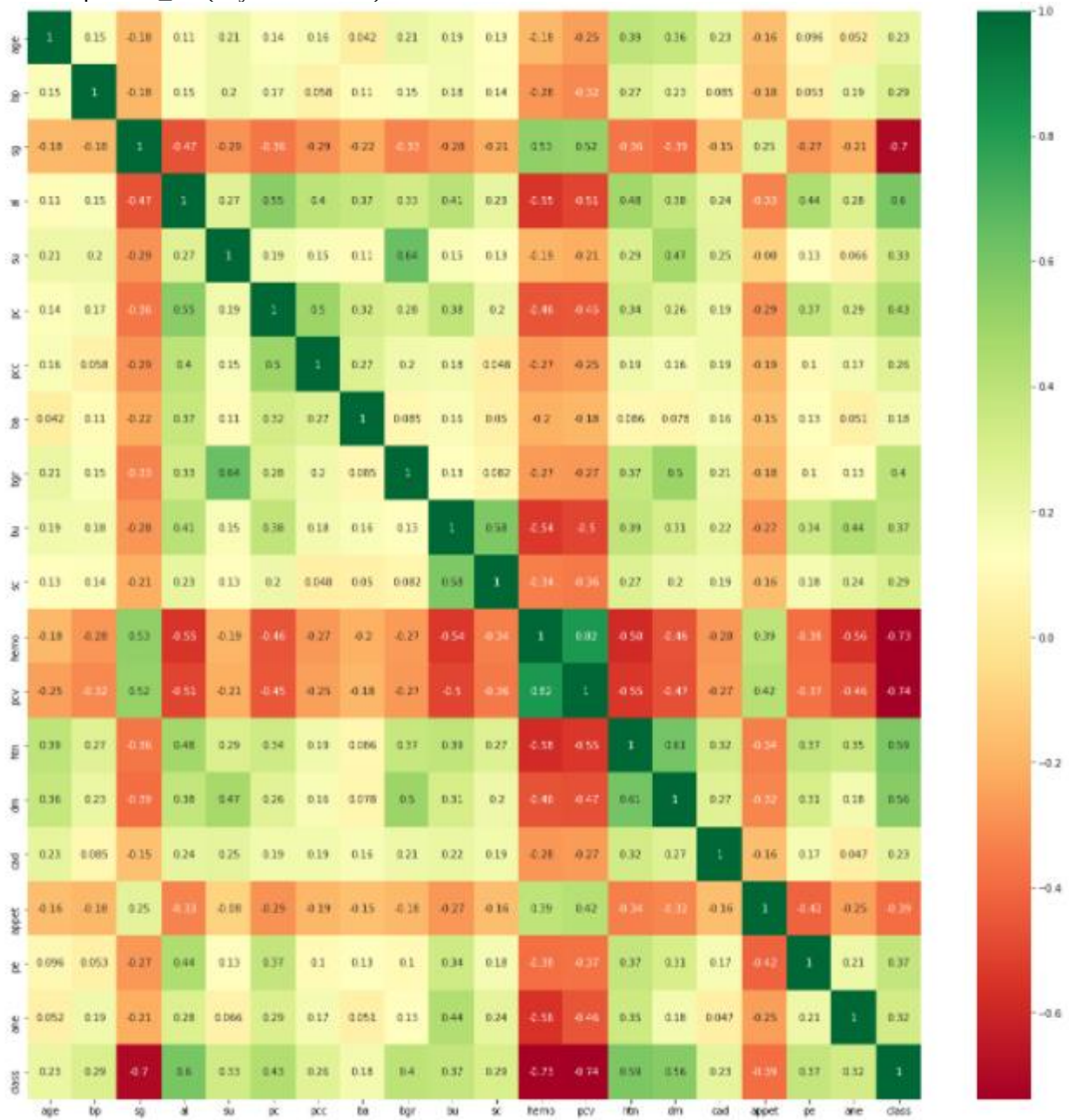


Fig 3.3.1: Co-relation matrix for CKD

Discretization

Conversion of a continuous-valued variable to a discrete value by creating a set of contiguous intervals that falls in the range of the variable's values is called discretization. We have categorized the using the following table. A classification algorithm such as Random Forests (RF) is chosen for its ability to handle high- dimensional data, benefits from discretization in the analysis of genetic data. Decision Tree is able to deal with both continuous and categorical attributes for classification. However, this technique handles the continuous attributes by discretization. A simple probabilistic classification algorithm i.e., Naive Bayes (NB) benefits from discretization of data in-order to perform well in many domains. But when we are using KNN since our data contains both continuous and categorical attributes we should use dummies.

Parameters	Measurement	Missing	Percent
Glucose	Num. (mg/dL)	44	11
Urea	Num. (mg/dL)	19	4.8
Creatinine	Num. (mg/dL)	17	4.3
Sodium	Num. (mEq/L)	87	21.8
Potassium	Num. (mmol/L)	88	22
Haemoglobin	Num. (g/dL)	52	13
Packed Cell Volume	Num.	71	17.8
White Blood Cell Count	Num. (cells/mcL)	106	26.5
Red Blood Cell Count	Num. (m.c./mcL)	131	32.8
Specific Gravity	Num. (1.002-1.030)	47	11.8
Urine Glucose	Category (0-5)	49	12.3
Albumin	Category (0-5)	46	11.5
Bacteria	Binary	4	1
Red Blood Cells in Urine	Binary	152	38
Pus Cell	Binary	65	16.25
Age	Num. (years)	9	2.3
Hypertension	Binary	2	0.5
Blood Pressure	Num. (mm/Hg)	12	3
Diabetes	Binary	2	0.5
Coronary Artery Disease	Binary	2	0.5
Appetite	Binary	1	0.25
Pedal edema	Binary	1	0.25
Anemia	Binary	1	0.25

Fig 3.3.2: class labels and attributes

Feature scaling

It is nothing but data normalization in data processing used to standardize the range of independent variables. This feature is very useful in data pre-processing step.

The Standard Scaler will assume that the data is normally distributed with in each attribute and will scale them in such a way that the distribution is now centred on 0, with a standard deviation is of 1.

The mean and standard deviation are calculated for the feature and then the feature is scaled based on:

$$y_i = \frac{y_i - \text{mean}(y)}{\text{stdev}(y)}$$

y_i represents the values of attribute y

Correlation coefficient method

We can find dependency between two attributes p and q using Correlation coefficient method using the formula.

$$r_{p,q} = \frac{\sum (p_i - \bar{p})(q_i - \bar{q})}{n \sigma_p \sigma_q} = \frac{\sum (p_i q_i) - n \bar{p} \bar{q}}{n \sigma_p \sigma_q}$$

n is the total number of patterns, p_i and q_i are respective values of p and q attributes in patterns i, $\bar{p}$ and $\bar{q}$ are respective mean values of p and q attributes, σ_p, σ_q are respective standard deviations values of p and q attributes. Generally, $-1 \leq r_{p,q} \leq +1$. If $r_{p,q} < 0$, then p and q are negatively correlated. If $r_{p,q} = 0$, then p and q are independent attributes and there is no correlation between them. If $r_{p,q} > 0$, then p and q are positively correlated.

We can drop the attributes that are having correlation coefficient value as 0 as it indicates that the variables are independent with respect to the prediction attribute.

In-order to choose n value for KNN algorithm we use the error rate. We choose the k value where the error rate is low.

3.4 Confusion Matrix

A confusion matrix is a table that is often used to describe the performance of a classification model (or "classifier") on a set of test data for which the true values are known.

		Predicted class	
		<i>P</i>	<i>N</i>
Actual Class	<i>P</i>	True Positives (TP)	False Negatives (FN)
	<i>N</i>	False Positives (FP)	True Negatives (TN)

Fig 3.4.1: format of confusion matrix

A true positive(tp) is a result where the model predicts the positive class correctly. Similarly, a true negative(tn) is an outcome where the model correctly predicts the negative class.

A false positive (fp) is an outcome where the model incorrectly predicts the positive class. And a false negative (fn) is an outcome where the model incorrectly predicts the negative class.

Sensitivity or Recall or hit rate or true positive rate (TPR)

It is the proportion of individuals who actually have the disease were identified as having the disease.

$$TPR = tp / (tp + fn)$$

Specificity, selectivity or true negative rate (TNR)

It is the proportion of individuals who actually do not have the disease were identified as not having the disease.

$$TNR = tn / (tn + fp) = 1 - FPR$$

Precision or positive predictive value (PPV)

If the test result is positive what is the probability that the patient actually has the disease. $PPV = tp / (tp + fp)$

Negative predictive value (NPV)

If the test result is negative what is the probability that the patient does not have disease.

$$NPV = tn / (tn + fn)$$

Miss rate or false negative rate (FNR)

It is the proportion of the individuals with a known positive condition for which the test result is negative.

$$FNR = fn / (fp + tn)$$

Fall-out or false positive rate (FPR)

It is the proportion of all the people who do not have the disease who will be identified as having the disease.

$$FPR = fp / (fp + tn)$$

False discovery rate (FDR)

It is the proportion of all the people identified as having the disease who do not have the disease.

$$FDR = fp / (fp + tp)$$

False omission rate (FOR)

It is the proportion of the individuals with a negative test result for which the true condition is positive.

$$FOR = fn / (fn + tn)$$

Accuracy

The accuracy reflects the total proportion of individuals that are correctly classified.

$$ACC = (tp + tn) / (tp + tn + fp + fn)$$

F1 score

It is the harmonic mean of precision and sensitivity

$$F1 = 2tp / (2tp + fp + fn)$$

4. Design

The design phase in machine learning includes from the preprocessing of the dataset then applying division on the dataset to training and test datasets. Applying the algorithms to fill the missing values. Applying the algorithms to predict the output and finding the best algorithm and conducting the tests on the system by using test dataset. The following figure specifies the various phases in designing the system. In this we divided 400 record sets into training and test dataset and apply the algorithm.

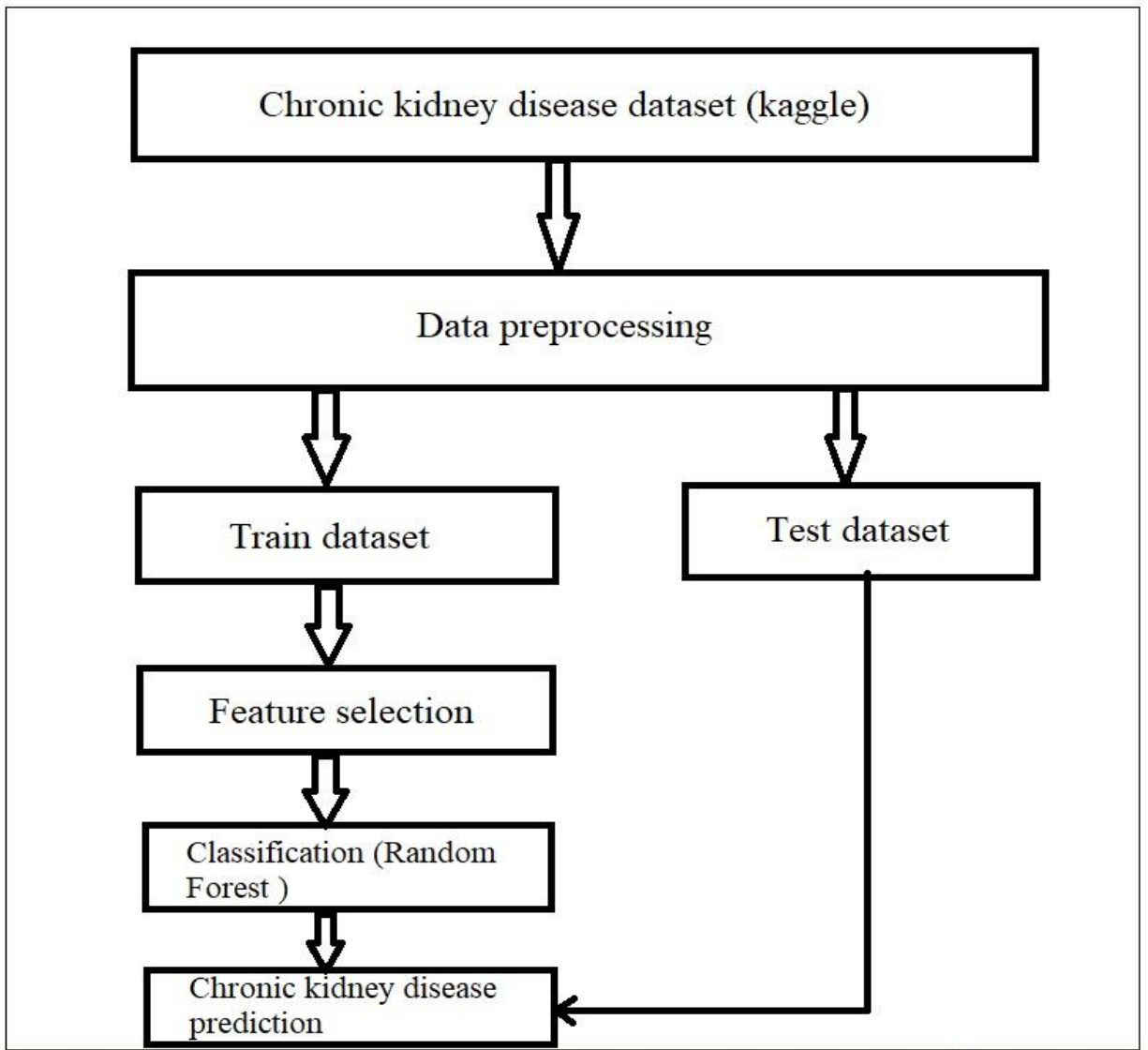


Fig 4.1: modeling the system

5. Implementation

Model file:

```
import numpy as np

import matplotlib.pyplot as plt import pandas as pd

import pickle

from sklearn.model_selection import train_test_split , GenericUnivariateSelect,chi2

import GridSearchCV from sklearn.preprocessing

import StandardScaler from sklearn.decomposition import PCA

from sklearn.feature_selection import SelectKBest

d=pd.read_csv('ckd.csv')

nullpercent=(d.isna().sum()/len(d))*100

d.drop(['rbc', 'sod','pot','wc','rc'], axis = 1,inplace=True)

nullpercent=(d.isna().sum()/len(d))*100

transformer = GenericUnivariateSelect(chi2, mode='k_best', param=20)

bestfeatures = SelectKBest(score_func=chi2, k=20)

transformer

bestfeatures

x = d.iloc[:, :-1] #independent columns
y = d.iloc[:, -1] #dependent or target column

Converting words to integer values:

d[['htn','dm','cad','pe','ane']] = d[['htn','dm','cad','pe','ane']].replace(to_replace={'yes':1,'no':0})

d[['pc']] = d[['pc']].replace(to_replace={'abnormal':1,'normal':0})

d[['pcc','ba']] = d[['pcc','ba']].replace(to_replace={'present':1,'notpresent':0})

d[['appet']] = d[['appet']].replace(to_replace={'good':1,'poor':0,'no':np.nan})

d['classification'] = d['classification'].replace(to_replace={'ckd':1.0,'ckd\t':1.0,'notckd':0.0,'no':0.0})

d.rename(columns={'classification':'class'},inplace=True)

d['pe'] = d['pe'].replace(to_replace='good',value=0) # Not having pedal edema is good
```

```

d['appet'] = d['appet'].replace(to_replace='no',value=0)
d['cad'] = d['cad'].replace(to_replace='tno',value=0)
d['dm'] = d['dm'].replace(to_replace={'\tno':0,\tyes':1,' yes':1, ":np.nan"})
d.drop('id',axis=1,inplace=True)
d['al'].fillna(d.al.mean(),inplace= True)
d['sg'].fillna(d.sg.mean(),inplace= True)
d['hemo'].fillna(d.hemo.mean(),inplace= True)
d['sc'].fillna(d.sc.mean(),inplace= True)
d['bu'].fillna(d.bu.mean(),inplace= True)
d['age'].fillna(d.age.mean(),inplace= True)
d['su'].fillna(d.su.mean(),inplace= True)
d['bp'].fillna(d.bp.mean(),inplace= True)
d['bgr'].fillna(d.bgr.mean(),inplace= True)
d['pcv'].ffill(inplace = True)
d['pcv'] =
d['pcv'].replace(to_replace={'\t?':d['pcv'].ffill(inplace=True),'\t43':d['pcv'].ffill(inplace=True)})
d['pcv'].unique()
d = d.astype({"pcv":'float'})
d['pc'].fillna(d.pc.mean(),inplace= True)
d['pcc'].fillna(d.pcc.mean(),inplace= True)
d['ba'].fillna(d.ba.mean(),inplace= True)
d['pcv'].fillna(d.pcv.median(),inplace= True)
d['htn'].fillna(d.htn.mean(),inplace= True)
d['dm'].fillna(d.dm.mean(),inplace= True)
d['cad'].fillna(d.cad.mean(),inplace= True)
d['appet'].fillna(d.appet.mean(),inplace= True)
d['pe'].fillna(d.pe.mean(),inplace= True)
d['ane'].fillna(d.ane.mean(),inplace= True)
d.isna().sum()
d.dtypes

```

Splitting Training and Test Set:

```

From sklearn.linear_model import LinearRegression
from sklearn.naive_bayes import GaussianNB
from sklearn.tree import DecisionTreeClassifier

import pickle

```

```

regressor = GaussianNB()
#Fitting model with trainig data regressor.fit(X1_train,y1_train) # Saving model to disk
with open('ckdproj.pickle','wb') as f:
pickle.dump(rfclass,f)
# Loading model to compare the results
rfc_model=pickle.load(open('ckdproj.pickle','rb'))
rfc_model.predict([[ 1.025 , 0.0, 131.000000 , 1.1, 15.8 , 53.0 , 0.0 , 0.0]])[0]
#print(rfc_model.predict([[ 1.025 , 0.0, 131.000000 , 1.1, 15.8 , 53.0 , 0.0 , 0.0]])[0])

```

Flask file:

```

from flask import Flask,render_template, request
import pickle
app=Flask(__name__)

```

```

@app.route('/')
def home():
    return render_template('home.html')

```

```

#@app.route('/precautionpage')
def precautionpage():
    return render_template('precautionpage.html')

```

```

@app.route('/predictionn')
def predictionn():
    return render_template('predictionn.html')

```

```

@app.route('/destination')
def destination():
    return render_template('destination.html')

```

```

@app.route('/predict',methods=['POST'])
def predict():

```

```

    if request.method == 'POST':

```

```

        sg=request.form['sg']

```

```

hemo=request.form['hemo']
htn=request.form['htn']
dm=request.form['dm']
pcv=request.form['pcv']
al=request.form['al']
sc=request.form['sc']
bgr=request.form['bgr']

data=[[float(sg),float(hemo),float(htn),float(dm),float(pcv),float(al),float(sc),float(bgr)]]
print(data)
r=pickle.load(open('ckdproj.pickle','rb'))
prediction_res=r.predict(data)[0]
if(prediction_res==1):
    pred="Your Kidneys tends to fail"
else:
    pred="Your Kidneys are safe"

return render_template('destination.html',prediction=pred)

@app.route('/submit',methods=['POST'])
def submit():

    if request.method == 'POST':

        bpsys=request.form['bpsys']
        bpdia=request.form['bpdia']
        dm=request.form['dm']
        htn=request.form['htn']
        cvd=request.form['cvd']
        fam=request.form['fam']

        data=[str(bpsys),str(bpdia),str(dm),str(htn),str(cvd),str(fam)]
        print(data)
        for i in data:
            if (((int(data[0])>120 and int(data[1])>80 and int(data[2])!=1 ) or (int(data[3])!=1 ) or
(int(data[4])!=1 or int(data[5])!=1)):

```

```

        return render_template('predictionn.html')
    else:
        return render_template('precautionpage.html')

return render_template('home.html')

if __name__ == '__main__':
    app.run()

```

HTML files:

home.html

```

<!DOCTYPE html>
<html >
    <head>
        <style>
body{
    background-repeat:no-repeat;
    background-attached:fixed;
    background-size:cover;
    background-image:url({{ url_for('static', filename='kidney1.jpg') }})
}
        </style>

        <link rel="stylesheet" href="{{ url_for('static', filename='hc.css') }}">
        <meta name="viewport" content="width=device-width,initial-scale=1,shrink-to-fit=no">
        <link rel="stylesheet"
href="https://stackpath.bootstrapcdn.com/bootstrap/4.3.1/css/bootstrap.min.css" integrity="sha384-
ggOyR0iXCbMQv3Xipma34MD+dH/1fQ784/j6cY/iJTQUOhcWr7x9JvoRxT2MZw1T"
crossorigin="anonymous">
        <title>Chronic Kidney Disease Prediction</title>

    </head>
    <body>
    <header>
        <div class="container1">
            <marquee width="100%" direction="left" height="100px">
<h1 class="logo">Chronic Kidney Disease Prediction web app</h1>
            </marquee>
        </div>
    </header>

```

```

<div class="container text-center" >
  <form action="{ { url_for('submit') } }" method="POST">
    <h1 style="color: rgb(255, 0, 106)">Know your risk factor of Kidneys - Basic checkup</h1>
    <h3 style="color: #7a0505"> Provide your basic Information of</h3>
    <input type="text" name="bpsys" placeholder="Blood Pressure: Systolic mm Hg (upper
number) "><br>
    <input type="text" name="bpdia" placeholder="Blood Pressure: Diastolic mm Hg (lower
number) "><br>
    <input type="text" name="dm" placeholder="Diabetes Mellitus: 0 for No, 1 for type 1, 2 for
type 2)"><br>
    <input type="text" name="htn" placeholder="Hypertension: 1 for yes 0 for no"><br>
    <input type="text" name="cvd" placeholder="Cardio Vascular Disease: 1 for yes 0 for
no"><br>
    <input type="text" name="fam" placeholder="History of any Chronic Kidney Disease in your
family: 1 for yes 0 for no"><br>

    <button class='btn bt-success'><p style=" background-color:
#5eee60;">SUBMIT</p></button>
  </form>

```

```

<script src="https://code.jquery.com/jquery-3.5.1.slim.min.js" integrity="sha384-
DfXdz2htPH0lsSSs5nCTpuj/zy4C+OGpamoFVy38MVBnE+IbbVYUew+OrCXaRkfj"
crossorigin="anonymous"></script>
<script src="https://cdn.jsdelivr.net/npm/popper.js@1.16.1/dist/umd/popper.min.js"
integrity="sha384-
9/reFTGAw83EW2RDu2S0VKaIzap3H66lZH81PoYlFhbGU+6BZp6G7niu735Sk7lN"
crossorigin="anonymous"></script>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@4.5.3/dist/js/bootstrap.min.js"
integrity="sha384-
w1Q4orYjBQndcko6MimVbzyY0tgp4pWB4lZ7lr30WKz0vr/aWKhXdBNmNb5D92v7s"
crossorigin="anonymous"></script>
</body>
</html>

```

destination.html

```

<!DOCTYPE html>
<html >
  <head>
  <style>
  body{

```

```

background-repeat:no-repeat;
background-attached:fixed;
background-size:cover;
background-image:url({{ url_for('static', filename='kidneypre.jpg') }})
}

</style>

<link rel="stylesheet" href="{{ url_for('static', filename='hc.css') }}">
<meta name="viewport" content="width=device-width,initial-scale=1,shrink-to-fit=no">
<link rel="stylesheet"
href="https://stackpath.bootstrapcdn.com/bootstrap/4.3.1/css/bootstrap.min.css" integrity="sha384-
ggOyR0iXCbMQv3Xipma34MD+dH/1fQ784/j6cY/iJTQUOhcWr7x9JvoRxT2MZw1T"
crossorigin="anonymous">
<title>Chronic Kidney Disease Prediction</title>
</head>
<body>

<header>
<div class="container1">
<marquee width="100%" direction="left" height="100px">
<h1 class="logo">Chronic Kidney Disease Prediction web app</h1>
</marquee>
</div>
</header>
<div class="container text-center" >
<form action="{{ url_for('predict') }}" method="POST">

<input type="text" name="sg" placeholder="sg(specific gravity)"><br>
<input type="text" name="hemo" placeholder="hemo(haemoglobin)"><br>
<input type="text" name="htn" placeholder="htn(hypertension)"><br>
<input type="text" name="dm" placeholder="dm(diabetes mellitus)"><br>
<input type="text" name="pcv" placeholder="pcv(packed cell volume)"><br>
<input type="text" name="al" placeholder="al(albumin)"><br>
<input type="text" name="sc" placeholder="sc(serum creatinine)"><br>
<input type="text" name="bgr" placeholder="bgr(blood glucose random )"><br>

<button class='btn bt-success'><p style=" background-color:
#5eee60;">PREDICT</p></button>
</form>

```

```

<div class="row">
  {% if prediction %}
    <h6 style="text-align: center;
font-size: 5em;
color: rgb(231, 18, 18);">{{prediction}}</h6>
    {% endif %}
  </div>
</div>

<script src="https://code.jquery.com/jquery-3.5.1.slim.min.js" integrity="sha384-
DfXdz2htPH0lsSSs5nCTpuj/zy4C+OGpamoFVy38MVBnE+IbbVYUew+OrCXaRkfj"
crossorigin="anonymous"></script>
<script src="https://cdn.jsdelivr.net/npm/popper.js@1.16.1/dist/umd/popper.min.js"
integrity="sha384-
9/reFTGAw83EW2RDu2S0VKaIzap3H66lZH81PoYlFhbGU+6BZp6G7niu735Sk7lN"
crossorigin="anonymous"></script>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@4.5.3/dist/js/bootstrap.min.js"
integrity="sha384-
w1Q4orYjBQndcko6MimVbzyY0tgp4pWB4lZ7lr30WKz0vr/aWKhXdBNmNb5D92v7s"
crossorigin="anonymous"></script>
</body>
</html>

```

6.Result Analysis

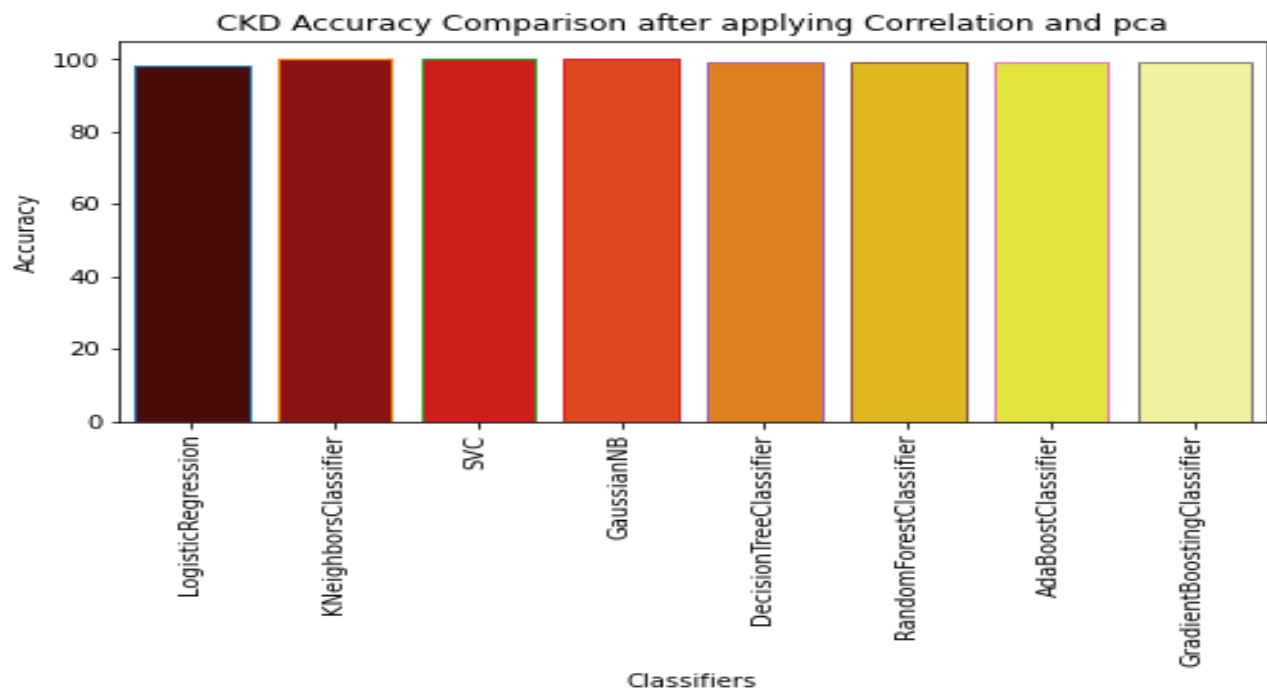


Fig 6.1: Accuracy after applying PCA and Co-relation

7. SCREENSHOTS

HOME PAGE:

CHRONIC KIDNEY DISEASE PREDICTION WEB APP

Know your risk factor of Kidneys - Basic checkup

Provide your basic Information of

Blood Pressure: Systolic mm Hg (upper number)

Blood Pressure: Diastolic mm Hg (lower number)

Diabetes Mellitus: 0 for No, 1 for type 1, 2 for type 2)

Hypertension: 1 for yes 0 for no

Cardio Vascular Disease: 1 for yes 0 for no

History of any Chronic Kidney Disease in your family: 1 for yes 0 for no

SUBMIT

Fig.7.1: Output screen for Home page

REPORT PAGE BASED ON RISK RESULT OF BASIC CHECKUP -1:

These might be the reasons that brings you here

1. Your blood pressure reading is high, you have stage 1 hypertension. Healthy kidneys produce a hormone called aldosterone to help the body regulate blood pressure. Kidney damage and uncontrolled high blood pressure each contribute to a negative spiral. As more arteries become blocked and stop functioning, the kidneys eventually fail.

2. DM 1, Type 1 diabetes is a chronic disease. That affecting about one-third of the patients

3. Hypertension, a global public health problem, is currently the leading factor in the global burden of disease. It is the major modifiable risk factor for heart disease, stroke and kidney failure. Chronic kidney disease (CKD) is both a common cause of hypertension and CKD is also a complication of uncontrolled hypertension.

4. Cardiovascular disease (CVD), the leading cause of death, is mostly precipitated by cardiometabolic risk and chronic kidney disease (CKD). CVD and kidney disease are closely interrelated and disease of one organ cause dysfunction of the other, ultimately leading to the failure of both organs.

5. Your family members having Kidney Disease so there are chances that may effect you as this is a Chronical disease.

please make sure of tests of

1. Haemoglobin(hemo)
2. Specific gravity(sg)
3. hypertension(htn)
4. diabetes mellitus(dm)
5. packed cell volume(pcv)
6. albumin(al)
7. serum creatinine(sc)
8. blood glucose random(bgr)

which are the major tests that determine whether you are suffering from chronic kidney disease or not...by using this web app

Predict your CKD

Fig 7.2: report page based on the risk result of basic checkup

PREDICTION FORM

Chronic Kidney Disease Prediction

127.0.0.1:5000/destination

CHRONIC KIDNEY DISEASE PREDICTION WEB APP



PREDICT

Fig 7.3: prediction form to predict CKD

PREDICTED FORM OF NO CKD

Chronic Kidney Disease Prediction

127.0.0.1:5000/predict

CHRONIC KIDNEY DISEASE PREDICTION WEB APP



PREDICT

Your Kidneys are safe

Fig 7.4: predicted form of no CKD

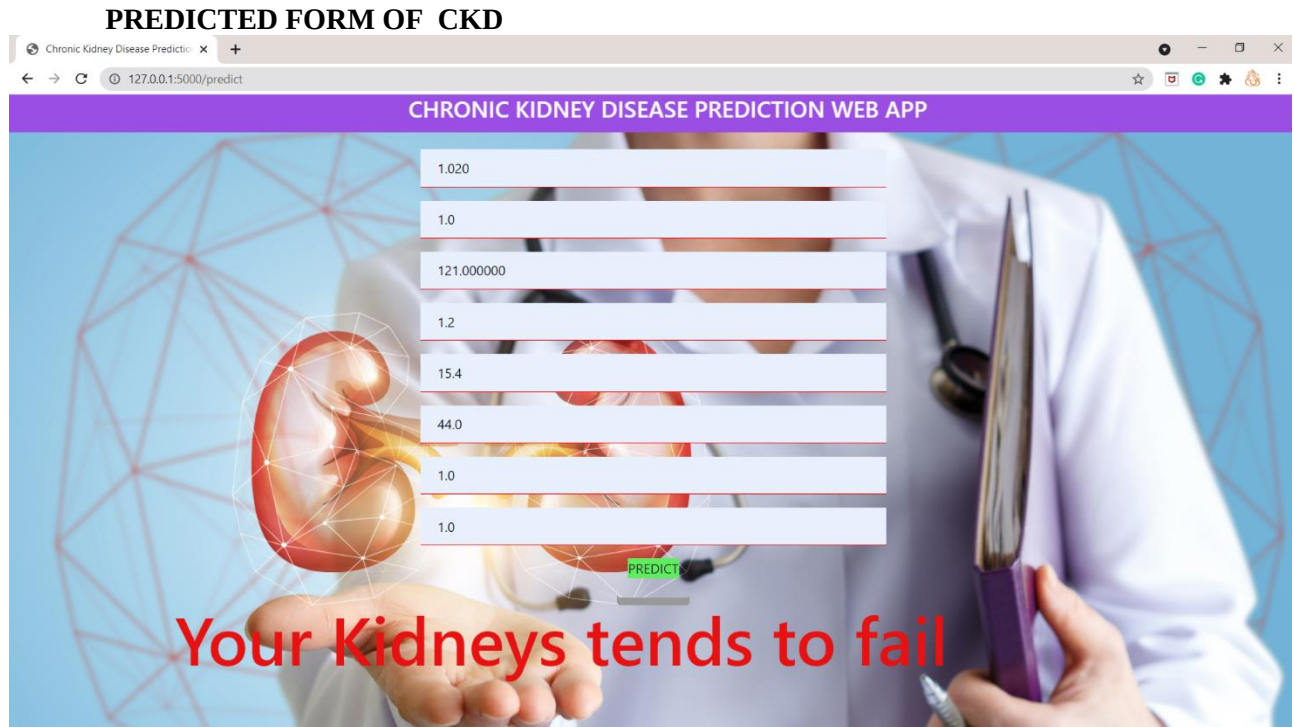


Fig 7.5: predicted form of CKD

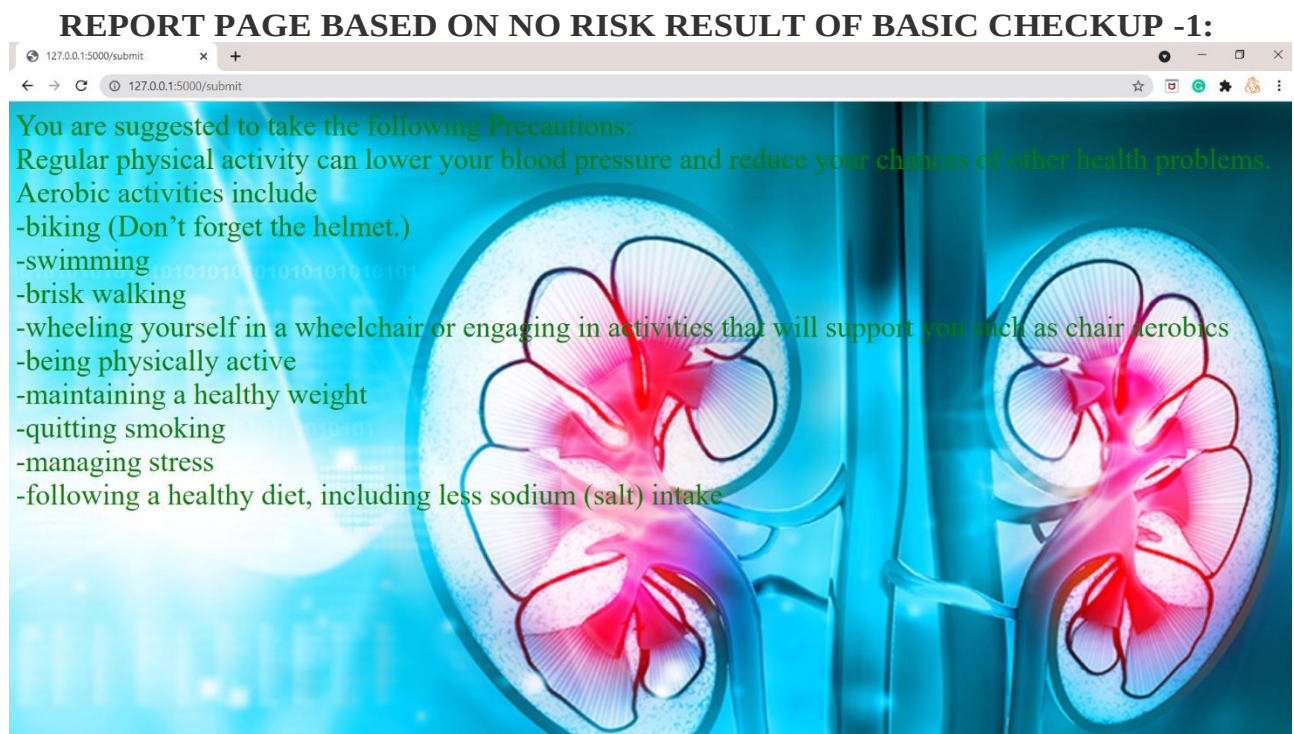


Fig 7.6: report page based on the no risk result of basic checkup

8. CONCLUSION

We have used 5 algorithms like Decision Trees, Random Forests, Naive Bayes, KNN and Logistic Regression in-order to predict presence or absence of chronic kidney disease. The accuracy varies for different algorithms. The accuracy for Decision tree algorithm is 98.92 . The accuracy for Random Forest algorithm is 98.73. The accuracy for NuSVC algorithm is 97.46. The accuracy for KNN algorithm is 94.93. The highest accuracy is given when we have used Logistic Regression algorithm using GaussianNB with an accuracy of 100 percent.

9. FUTURESCOPE

This project further can be developed as Android application to overcome the limitation of accessing the system by only desktop and also suggest them about their stage in chronic kidney diseases. And also suggest the required treatment that to be taken by the person to cure from the chronic kidney disease at that particular stage.

10. REFERENCES

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